Packet Tracer - TCP and UDP Communications

Objectives

Part 1: Generate Network Traffic in Simulation Mode

Part 2: Examine the Functionality of the TCP and UDP Protocols

Background

This simulation activity is intended to provide a foundation for understanding TCP and UDP in detail. Packet Tracer simulation mode provides you the ability to view the state of different PDUs as they travel through the network.

Packet Tracer Simulation mode enables you to view each of the protocols and the associated PDUs. The steps outlined below lead you through the process of requesting network services using various applications that are available on a client PC. You will explore the functionality of the TCP and UDP protocols, multiplexing, and the function of port numbers in determining which local application requested the data or is sending the data. Packet Tracer will not score this activity.

Instructions

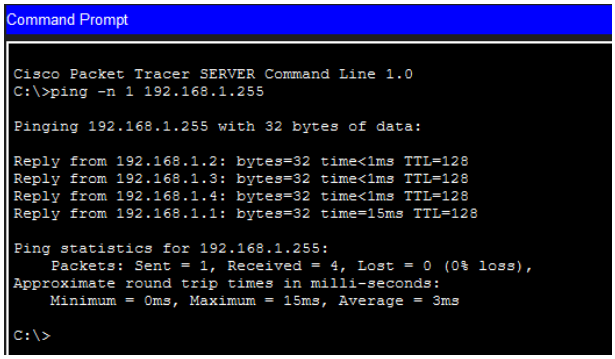
Part 1: Generate Network Traffic in Simulation Mode and View Multiplexing

Step 1: Generate traffic to populate Address Resolution Protocol (ARP) tables.

Perform the following task to reduce the amount of network traffic viewed in the simulation.

a. Click **MultiServer** and click the **Desktop** tab > **Command Prompt**.

b. Enter the **ping -n 1 192.168.1.255** command. You are pinging the broadcast address for the client LAN. The command option will send only one ping request rather than the usual four. This will take a few seconds as every device on the network responds to the ping request from MultiServer.



c. Close the **MultiServer** window.

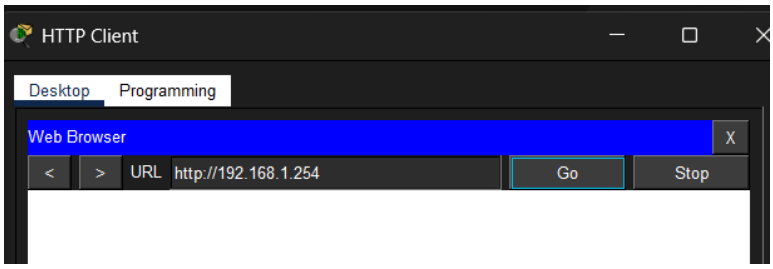
Step 2: Generate web (HTTP) traffic.

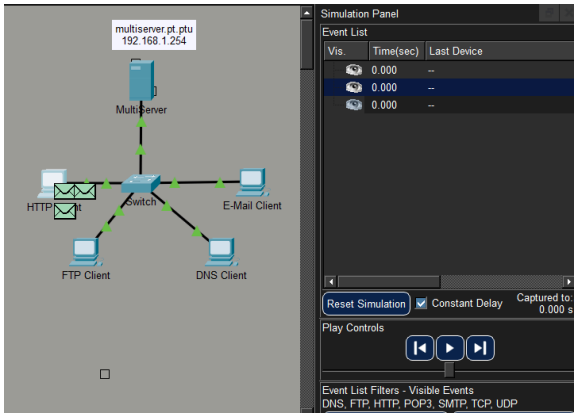
a. Switch to Simulation mode.

b. Click **HTTP Client** and open the **Web Browser** from the desktop.

c. In the URL field, enter **192.168.1.254** and click **Go**. Envelopes (PDUs) will appear in the topology window.

d. Minimize, but do not close, the **HTTP Client** configuration window.



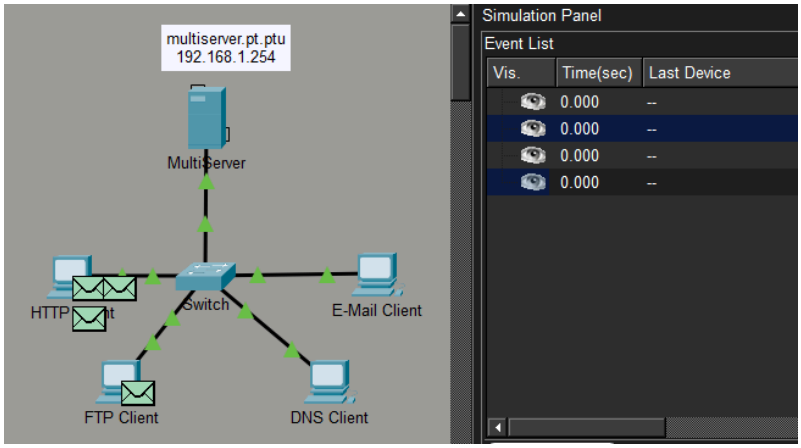


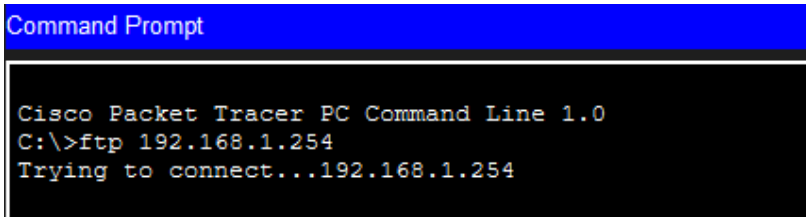
Step 3: Generate FTP traffic.

a. Click **FTP Client** and open the **Command Prompt** from the desktop

b. Enter the **ftp 192.168.1.254** command. PDUs will appear in the simulation window.

c. Minimize, but do not close, the **FTP Client** configuration window.



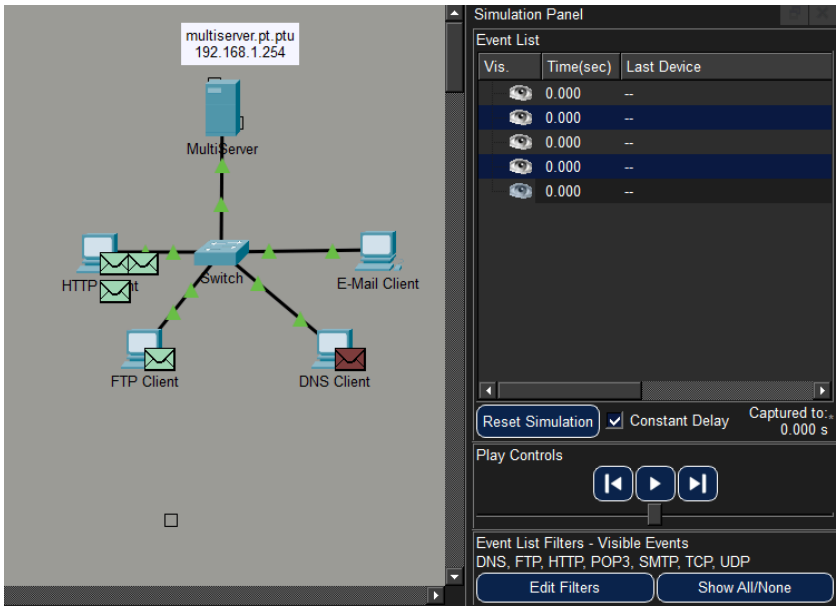


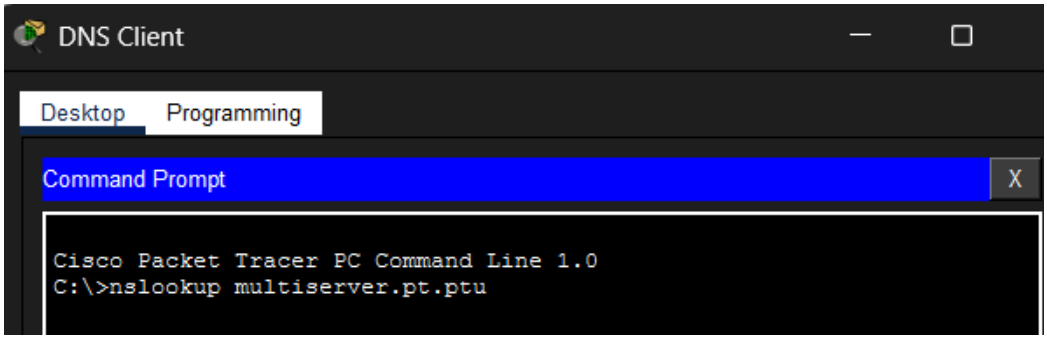
Step 4: Generate DNS traffic.

a. Click DNS Client and open the **Command Prompt**.

b. Enter the **nslookup multiserver.pt.ptu** command. A PDU will appear in the simulation window.

c. Minimize, but do not close, the **DNS Client** configuration window.





Step 5: Generate Email traffic.

a. Click **E-Mail Client** and open the **E Mail** tool from the Desktop.

b. Click **Compose** and enter the following information:

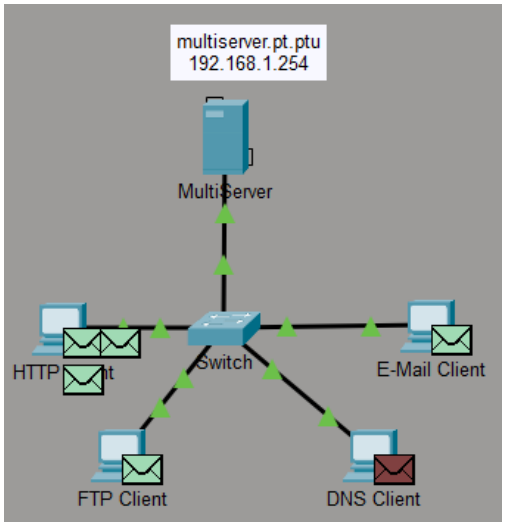
   1) **To:** user@multiserver.pt.ptu

   2) **Subject:** personalize the subject line

   3) **E-Mail Body:** personalize the Email

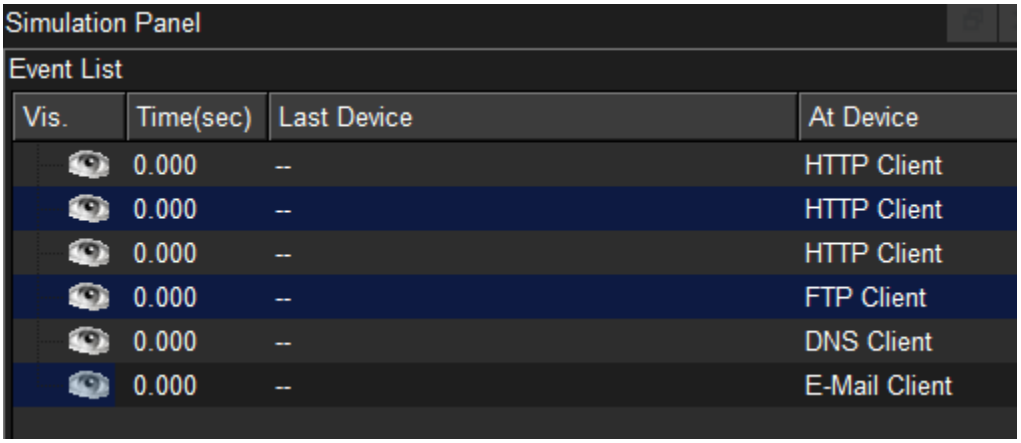
c. Click **Send**.

d. Minimize, but do not close, the **E-Mail Client** configuration window.



Step 6: Verify that the traffic is generated and ready for simulation.

There should now be PDU entries in the simulation panel for each of the client computers.



Simulation Panel

Event List

Vis.	Time(sec)	Last Device	At Device
	0.000	--	HTTP Client
	0.000	--	HTTP Client
	0.000	--	HTTP Client
	0.000	--	FTP Client
	0.000	--	DNS Client
	0.000	--	E-Mail Client

Step 7: Examine multiplexing as the traffic crosses the network.

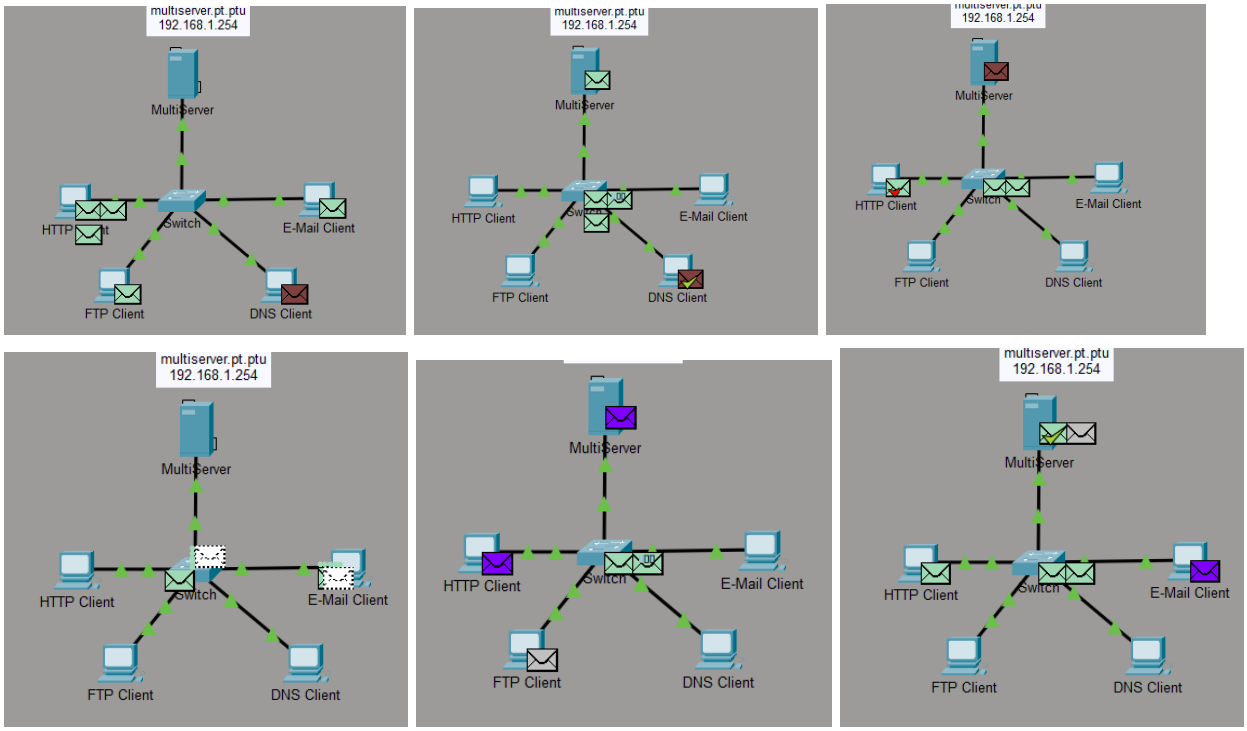
You will now use the **Capture/Forward button** in the Simulation Panel to observe the different protocols travelling on the network.

Note: The **Capture/Forward** button ‘ **>|** ‘ is a small arrow pointing to the right with a vertical bar next to it.

a. Click **Capture/Forward** once. All of the PDUs travel to the switch.

b. Click **Capture/Forward** six times and watch the PDUs from the different hosts as they travel on the network. Note that only one PDU can cross a wire in each direction at any given time.

1. What is this called? **Multiplexing**
2. A variety of PDUs appears in the event list in the Simulation Panel. What is the meaning of the different colors? **แทนโปรโตคอลต่าง ๆ (HTTP, FTP, DNS, SMTP เป็นต้น)**



Part 2: Examine Functionality of the TCP and UDP Protocols

Step 1: Examine HTTP traffic as the clients communicate with the server.

a. Click **Reset Simulation**.

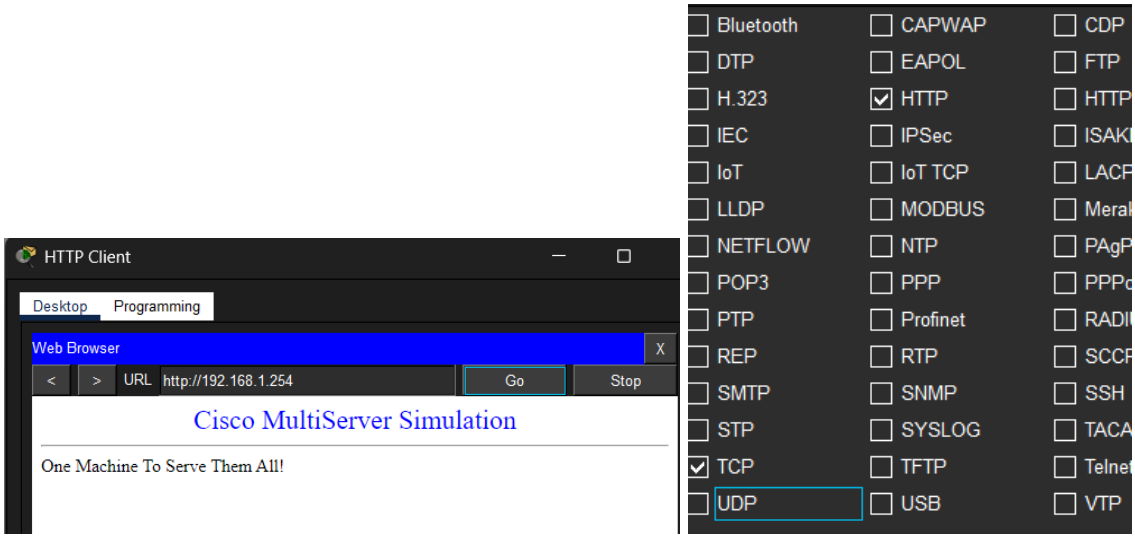
b. Filter the traffic that is currently displayed to only **HTTP** and **TCP** PDUs. To filter the traffic that is currently displayed:

1) Click **Edit Filters** and toggle the **Show All/None** button.

2) Select **HTTP** and **TCP**. Click the red “x” in the upper right-hand corner of the Edit Filters box to close it. Visible Events should now display only **HTTP** and **TCP** PDUs.

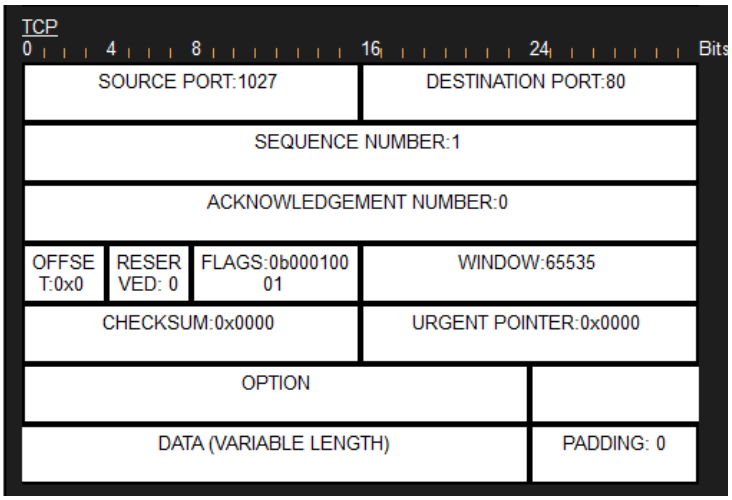
c. Open the browser on HTTP Client and enter **192.168.1.254** in the URL field. Click **Go** to connect to the server over HTTP. Minimize the HTTP Client window.

d. Click **Capture/Forward** until you see a PDU appear for HTTP. Note that the color of the envelope in the topology window matches the color code for the HTTP PDU in the Simulation Panel.



Why did it take so long for the HTTP PDU to appear? **เพราะต้องทำ TCP 3-way handshake (SYN, SYN-ACK, ACK)**

e. Click the PDU envelope to show the PDU details. Click the **Outbound PDU Details** tab and scroll down to the second to the last section.



is the section labeled?

ชื่อ section: **Transmission Control Protocol**

Reliable ไหม: **ใช่ (Reliable)**
จดค่า:

- SRC PORT
- DEST PORT
- SEQ
- ACK

Are these communications considered to be reliable?

Record the **SRC PORT**, **DEST PORT**, **SEQUENCE NUM**, and **ACK NUM** values.

f. Look at the value in the Flags field, which is located next to the Window field. The values to the right of the “b” represent the TCP flags that are set for this stage of the data conversation. Each of the six places corresponds to a flag. The presence of a “1” in any place indicates that the flag is set. More than one flag can be set at a time. The values for the flags are shown below.

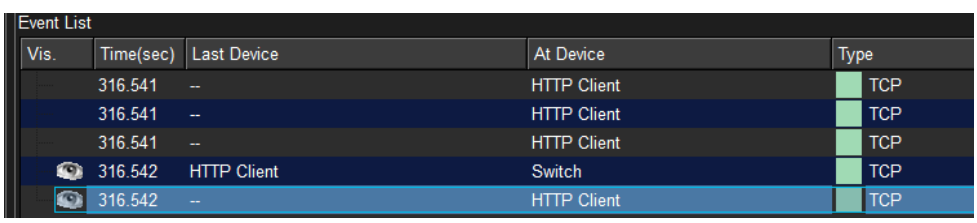
Flag Place	**6**	**5**	**4**	**3**	**2**	**1**
Value	URG	ACK	PSH	RST	SYN	FIN

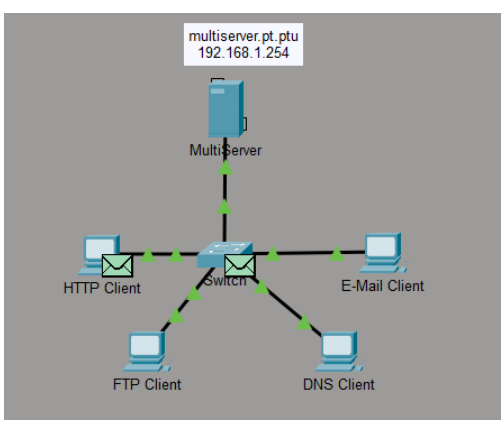
Which TCP flags are set in this PDU?

: SYN flag is set (SYN = 1)

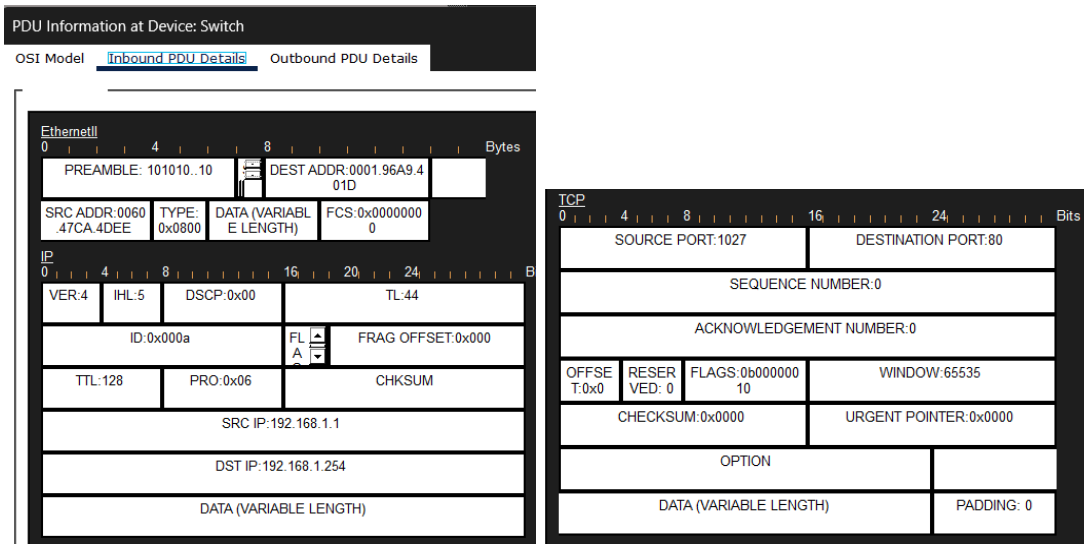
เป็น PDU แรกของ TCP เพื่อขอเริ่มการเชื่อมต่อกับ server

g. Close the PDU and click **Capture/Forward** until a PDU with a checkmark returns to the **HTTP Client**.



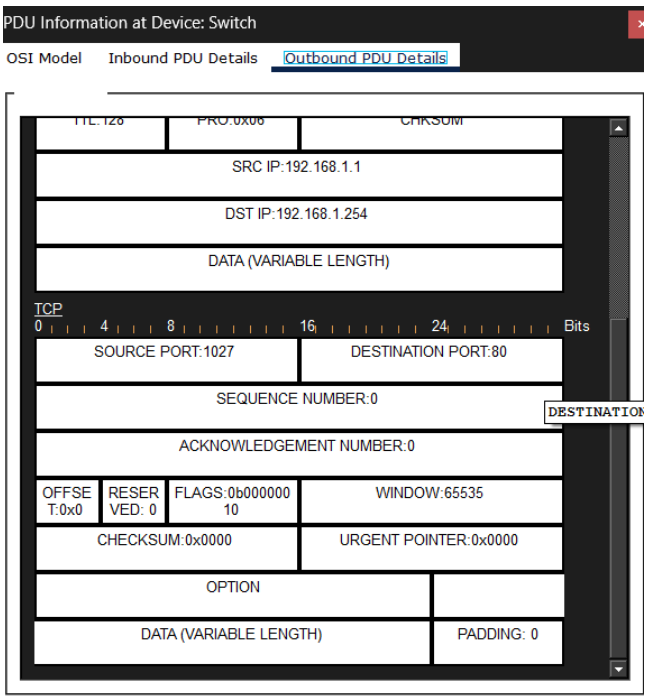


h. Click the PDU envelope and select **Inbound PDU Details**.



How are the port and sequence numbers different than before? **SRC ↔ DEST สลับกัน / ACK เพิ่มค่า**

i. Click the HTTP PDU which **HTTP Client** has prepared to send to **MultiServer**. This is the beginning of the HTTP communication. Click this second PDU envelope and select **Outbound PDU Details**.

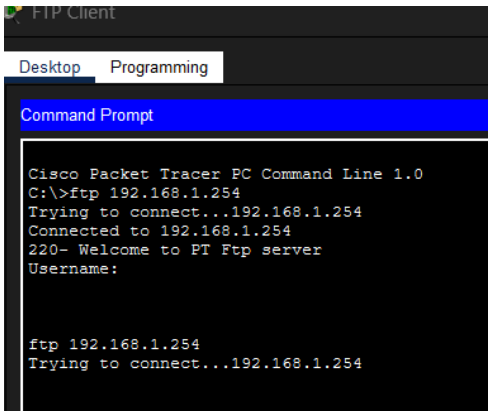


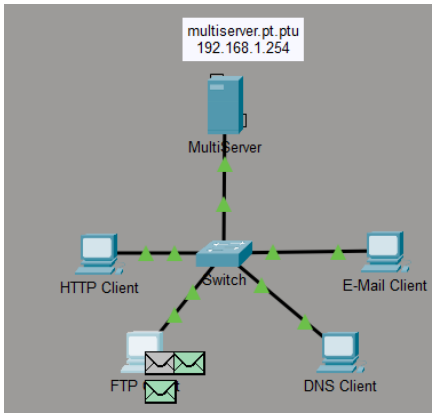
What information is now listed in the TCP section? How are the port and sequence numbers different from the previous two PDUs?

j. Reset the simulation.

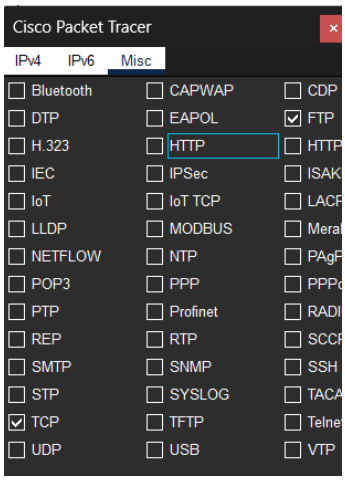
Step 2: Examine FTP traffic as the clients communicate with the server.

a. Open the command prompt on the FTP Client desktop. Initiate an FTP connection by entering **ftp 192.168.1.254**.



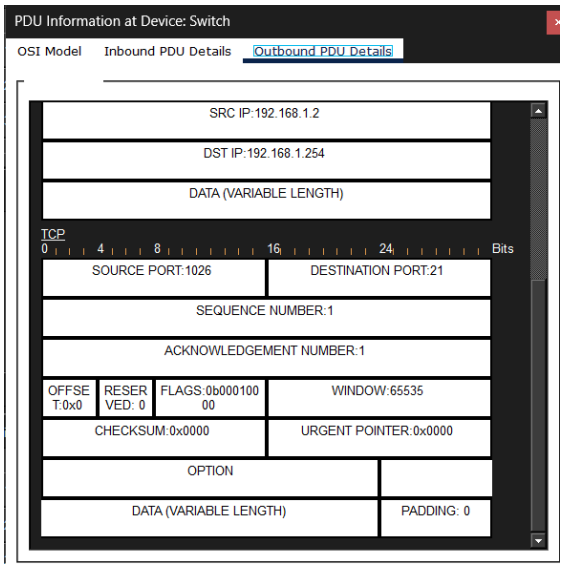


b. In the Simulation Panel, change **Edit Filters** to display only **FTP** and **TCP**.



c. Click **Capture/Forward**. Click the second PDU envelope to open it.

Click the **Outbound PDU Details** tab and scroll down to the TCP section.



Are these communications considered to be reliable?

เป็นการสื่อสารแบบ Reliable เนื่องจาก FTP ใช้ TCP ซึ่งมีการยืนยันการรับส่งข้อมูล

d. Record the **SRC PORT**, **DEST PORT**, **SEQUENCE NUM**, and **ACK NUM** values.

- SRC PORT: 1026

- DEST PORT: 21

- SEQUENCE NUM: 0

- ACK NUM: 0

What is the value in the flag field?

Flag มักเป็น: SYN (หรือ SYN + ACK ถ้าเป็นฝั่ง server)

e. Close the PDU and click **Capture/Forward** until a PDU returns to the **FTP Client** with a checkmark.

Vis.	Time(sec)	Last Device	At Device	Type
	0.002	Switch	MultiServer	FTP
	0.002	--	FTP Client	TCP
	0.002	--	MultiServer	FTP
	0.003	FTP Client	Switch	TCP
	0.003	Switch	MultiServer	TCP
	0.003	MultiServer	Switch	FTP
	0.004	Switch	MultiServer	TCP
	0.004	MultiServer	Switch	TCP
	0.004	Switch	FTP Client	FTP
	0.005	MultiServer	Switch	TCP
	0.005	Switch	FTP Client	TCP
	0.006	Switch	FTP Client	TCP
	0.007	FTP Client	Switch	TCP
	0.008	Switch	MultiServer	TCP
	0.008	--	MultiServer	FTP
	0.009	MultiServer	Switch	FTP
	0.010	Switch	FTP Client	FTP
	0.088	--	FTP Client	TCP
	0.089	FTP Client	Switch	TCP
Visible	0.090	Switch	MultiServer	TCP

f. Click the PDU envelope and select **Inbound PDU Details**.

How are the port and sequence numbers different than before?

สิ่งที่เปลี่ยน

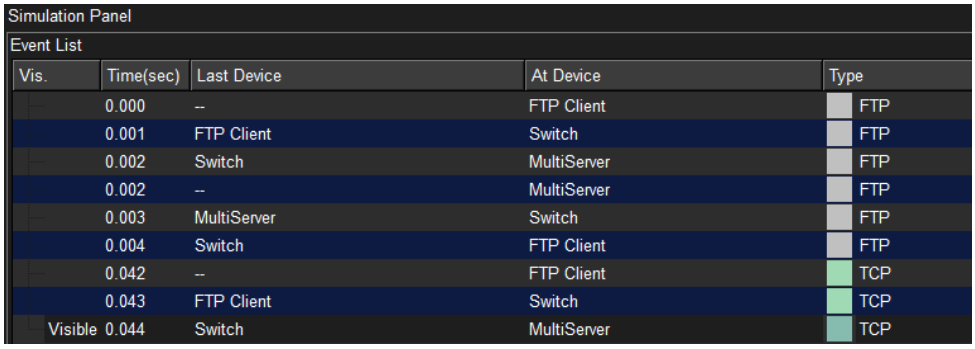
- **Source Port ↔ Destination Port สลับกัน**
- **Sequence Number เปลี่ยน** (เดินต่อจากเดิม)
- **ACK Number เพิ่มขึ้น** เพื่อยืนยันข้อมูลที่รับ

g. Click the **Outbound PDU Details** tab.

How are the port and sequence numbers different from the previous results?

- **Port ยังเป็นคู่เดิม (Client ↔ Server)**
- **Sequence Number เดินต่อไป**
- **แสดงว่า TCP connection ยังดำเนินอยู่**

h. Close the PDU and click **Capture/Forward** until a second PDU returns to the **FTP Client**. The PDU is a different color.

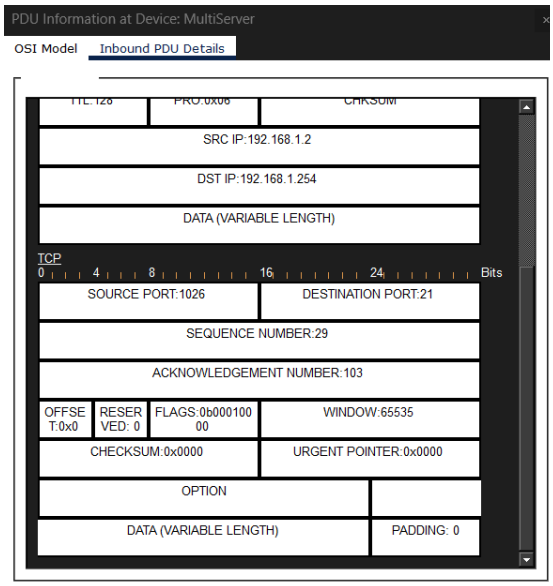


Simulation Panel

Event List

Vis.	Time(sec)	Last Device	At Device	Type
	0.000	--	FTP Client	FTP
	0.001	FTP Client	Switch	FTP
	0.002	Switch	MultiServer	FTP
	0.002	--	MultiServer	FTP
	0.003	MultiServer	Switch	FTP
	0.004	Switch	FTP Client	FTP
	0.042	--	FTP Client	TCP
	0.043	FTP Client	Switch	TCP
Visible	0.044	Switch	MultiServer	TCP

i. Open the PDU and select **Inbound PDU Details**. Scroll down past the TCP section.



What is the message from the server?

220 Service ready for new user

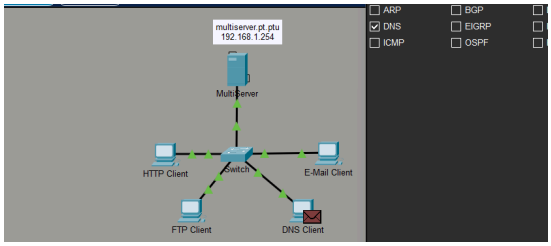
j. Click Reset Simulation.

Step 3: Examine DNS traffic as the clients communicate with the server.

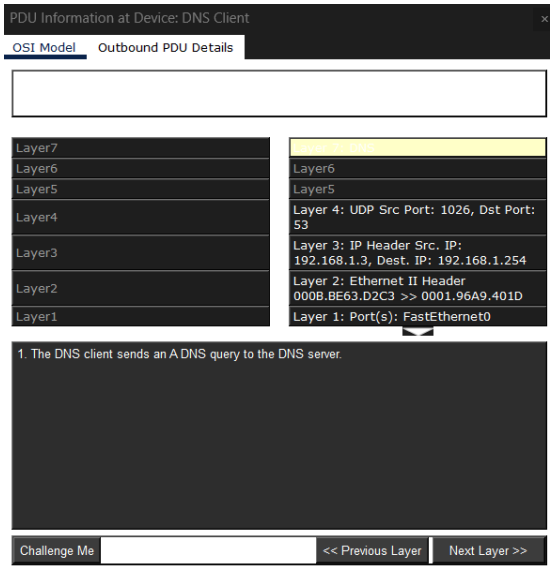
a. Repeat the steps in Part 1 to create DNS traffic.

b. In the Simulation Panel, change **Edit Filters** to display only **DNS** and **UDP**.

c. Click the PDU envelope to open it.



d. Look at the OSI Model details for the outbound PDU.



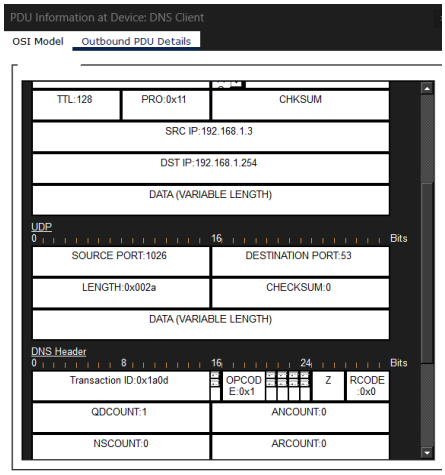
What is the Layer 4 protocol?

UDP (User Datagram Protocol)

Are these communications considered to be reliable?

ไม่เป็น Reliable เพราะ UDP ไม่มีการยืนยันการรับข้อมูล

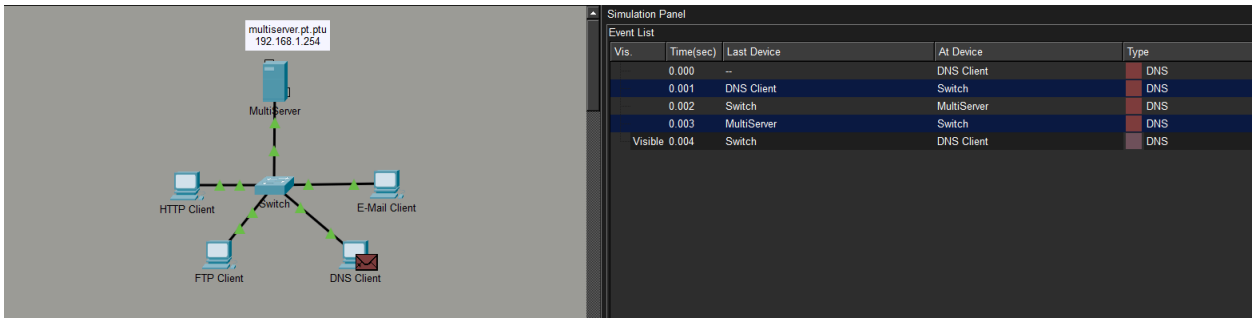
e. Open the Outbound PDU Details tab and find the UDP section of the PDU formats. Record the **SRC PORT** and **DEST PORT** values.



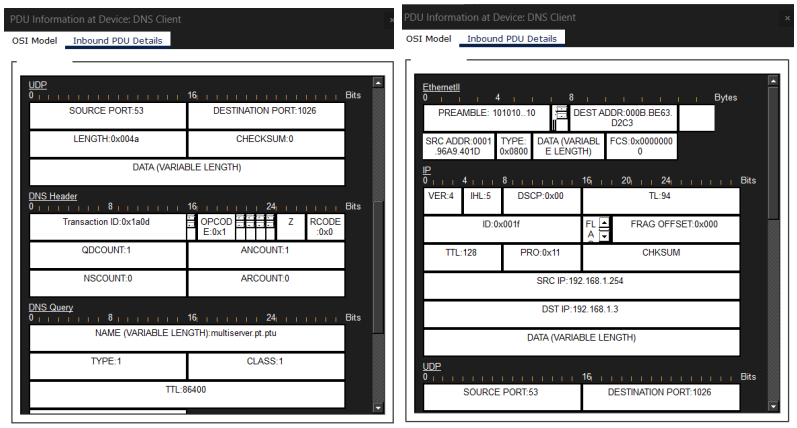
Why are there no sequence and acknowledgement numbers?

เพราะ UDP เป็นโปรโตคอลแบบไม่เชื่อมต่อ (connectionless) จึงไม่มีการใช้ sequence และ acknowledgment numbers

f. Close the **PDU** and click **Capture/Forward** until a PDU with a check mark returns to the **DNS Client**.



g. Click the PDU envelope and select **Inbound PDU Details**.



How are the port and sequence numbers different than before?

หมายเลขพอร์ตต้นทางและปลายทางจะ สลับกัน เมื่อเทียบกับตอนที่ส่งคำขอออกไป โดยพอร์ตต้นทางจะเป็นพอร์ตของ DNS Server (53) และพอร์ตปลายทางจะเป็นพอร์ตสุ่มของ DNS Client ส่วน หมายเลขลำดับ (Sequence Number) ยังคง ไม่มี ทั้งก่อนและหลัง เนื่องจาก DNS ใช้โปรโตคอล UDP ซึ่งไม่ใช้ sequence number และ acknowledgment number

What is the last section of the **PDU** called?

ส่วนสุดท้ายของ PDU เรียกว่า Domain Name System (DNS)

What is the IP address for the name **multiserver.pt.ptu**?

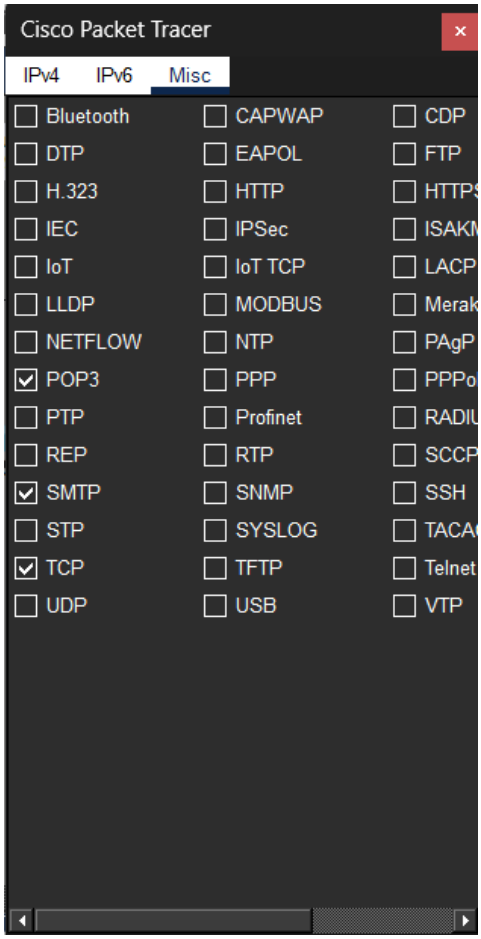
192.168.1.254

h. Click Reset Simulation.

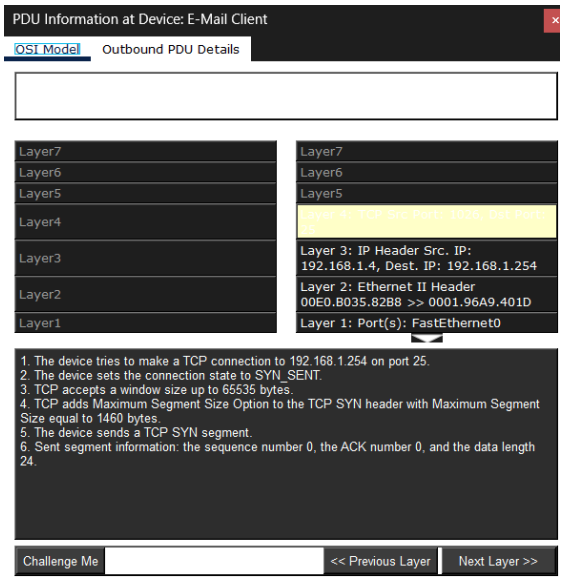
Step 4: Examine email traffic as the clients communicate with the server.

a. Repeat the steps in Part 1 to send an email to **user@multiserver.pt.ptu**.

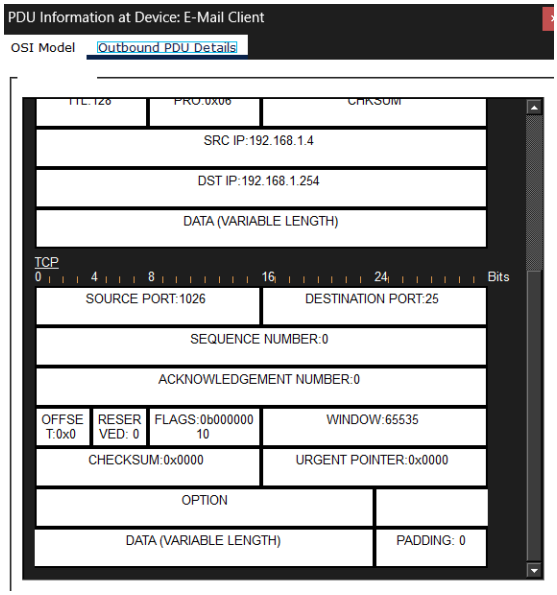
b. In the Simulation Panel, change **Edit Filters** to display only **POP3, SMTP** and **TCP**



c. Click the first PDU envelope to open it.



d. Click the **Outbound PDU Details** tab and scroll down to the last section.



What transport layer protocol does email traffic use?

ใช้โปรโตคอล **TCP**

Are these communications considered to be reliable?

เป็นการสื่อสารแบบ **Reliable** เพราะ TCP มีการยืนยันการรับส่งข้อมูล

e. Record the **SRC PORT**, **DEST PORT**, **SEQUENCE NUM**, and **ACK NUM** values. What is the flag field value?

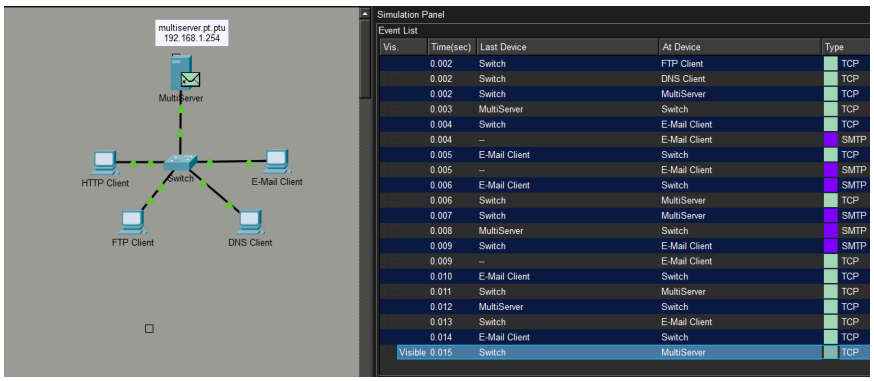
- SRC PORT: 1026

- DEST PORT: 25

- SEQUENCE NUM: 0

- ACK NUM: 0

f. Close the **PDU** and click **Capture/Forward** until a PDU returns to the **E-Mail Client** with a checkmark.

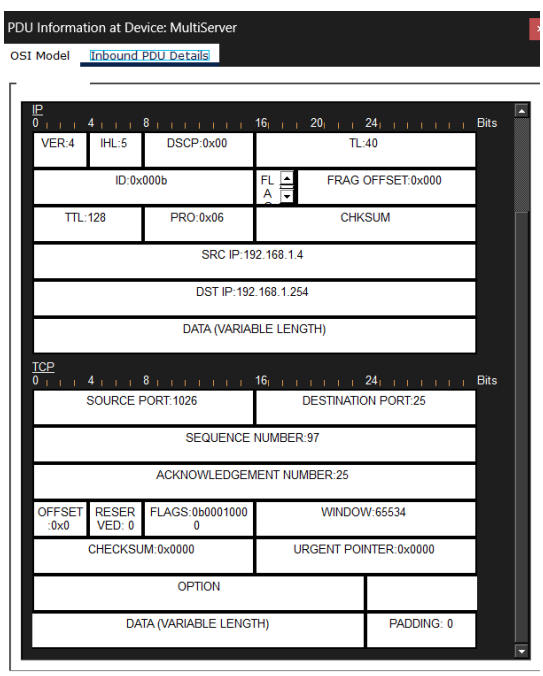


g. Click the TCP PDU envelope and select **Inbound PDU Details**.

How are the port and sequence numbers different than before?

พอร์ตต้นทางและปลายทาง **สลับกัน** เมื่อเทียบกับตอนส่งออก และค่า **sequence / acknowledgment numbers เปลี่ยนไป** เนื่องจากเป็นการตอบกลับจากเซิร์ฟเวอร์

h. Click the **Outbound PDU Details** tab.



How are the port and sequence numbers different from the previous two results?

หมายเลขพอร์ตยังคงเป็นคู่เดิมของการสื่อสาร (Client ↔ Server) แต่ค่า **sequence number เพิ่มขึ้น/เปลี่ยนไป** แสดงว่าการสื่อสารใน session เดิมยังดำเนินต่อ

i. There is a second **PDU** of a different color that **E-Mail Client** has prepared to send to **MultiServer**. This is the beginning of the email communication. Click this second PDU envelope and select **Outbound PDU Details**.

How are the port and sequence numbers different from the previous two **PDU**s?

พอร์ตยังคงเป็นพอร์ตของบริการอีเมลเดิม แต่ค่า **sequence number เพิ่มขึ้นมากกว่าเดิม** เนื่องจากมีการส่งข้อมูลอีเมลจริง

What email protocol is associated with TCP port 25?

SMTP (Simple Mail Transfer Protocol)

What protocol is associated with TCP port 110?

POP3 (Post Office Protocol version 3)